



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to MOOKAMBIKAI SERVICE STATION

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

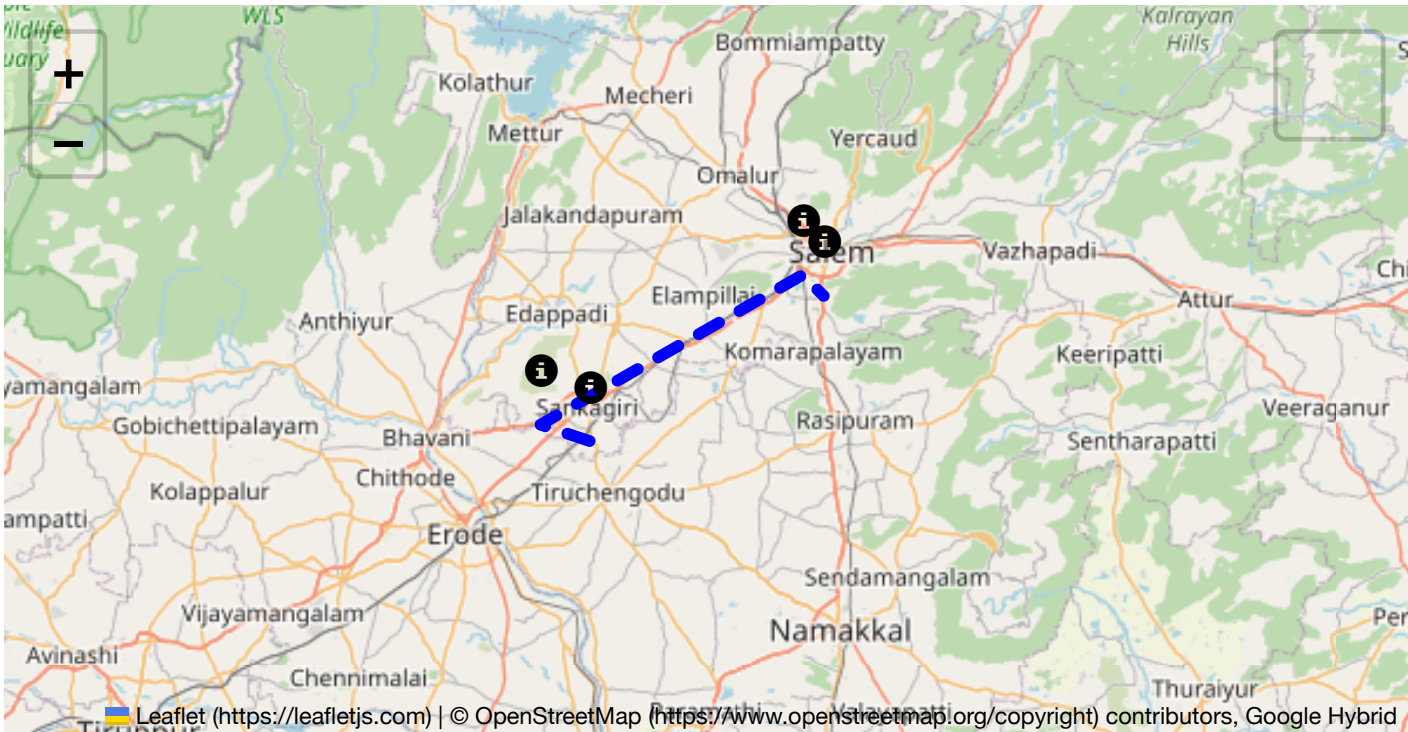
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.

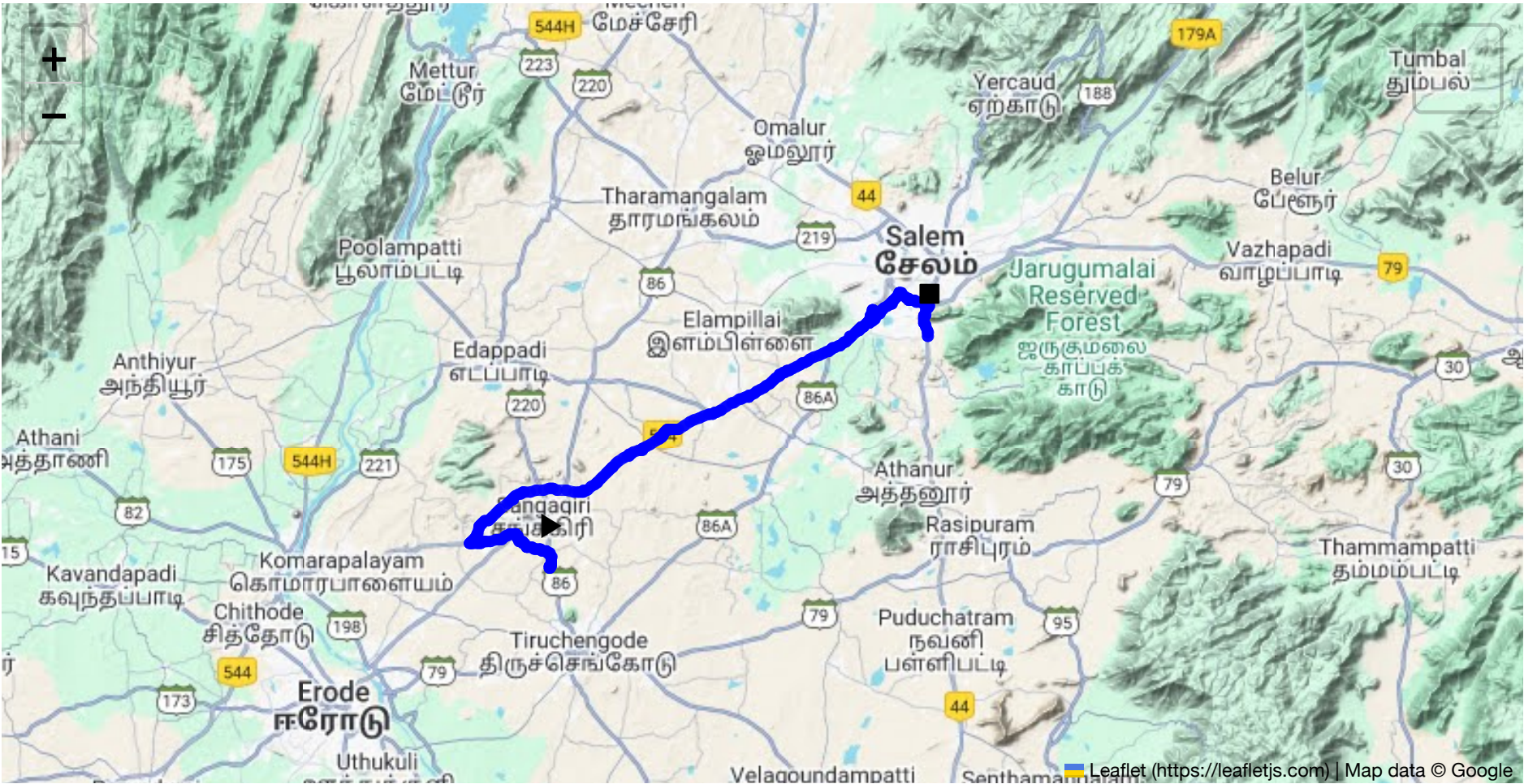


Route Summary:
Total Distance: 56.50 km
Estimated Duration: 1.1 hours

Adjusted Duration (Heavy Vehicle): 1.4 hours

Start: (11.4381, 77.8734)

End: (11.6028, 78.1475)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W in Sangagiri to Gajjalnaickenpatti in Salem, via FR48+7XQ in Padaiveedu and Kondalampatti in Salem, spans approximately 56.50 kilometers. It involves state highways and urban roads, typically requiring about 1.15 hours for heavy vehicles, including trucks carrying hazardous materials.

1. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate with hot summers, a monsoon season, and mild winters:

- **Summer (March to June):** Temperatures can be very high, and heatwaves may affect vehicle performance.
- **Monsoon (July to September):** Heavy rains could lead to waterlogging and slippery roads, potentially causing delays and increasing accident risk.
- **Winter (November to February):** Mild conditions prevail, though occasional fog in the early morning can reduce visibility.

2. Traffic Patterns and Congestion-Prone Areas

- **Peak Hours:** Both Sangagiri and Salem are busy during peak urban hours, typically between 8:00-10:00 AM and 5:00-8:00 PM.
- **Congestion-Prone Areas:** Urban centers in Salem, including traffic junctions and markets, often witness slow-moving traffic. Padaiveedu might have intermediate traffic due to its location on a busy

main road.

3. Road Quality and Infrastructure

- **Conditions:** State highways and urban roads are generally well-maintained. Expect rough patches near construction zones or under maintenance.
- **Infrastructure:** Adequate signage, but some rural areas might lack advanced signaling systems. Transport facilities in urban areas are developed.

4. Suggestions for Alternative Routes

In case of heavy congestion or roadblocks, consider:

- Using Tamil Nadu State Highways (e.g., NH44) if accessible, for quicker transit.
- Local bypass roads near congested urban segments to avoid traffic jams.

5. Local Regulations Affecting Hazardous Material Transport

- **Timings:** Transportation of hazardous materials is subject to restrictions, usually outside peak hours.
- **Permissions:** Specific permits and adherence to safety protocols are mandatory.
- **Escort:** Certain material may require police or additional security escort.

6. Overview of Historical Incidents

- Instances of accidents involving heavy vehicles in the region often relate to brake failure and road skidding during monsoons.
- No major hazardous material-specific incidents reported recently, though caution is advised.

7. Environmental Considerations and Sensitive Areas

- Areas around Salem and Tiruchengode are agriculturally intensive. Efforts should be made to minimize pollution and prevent spillage of hazardous materials.

8. Communication Coverage

- Mobile network coverage is generally good along state highways, with potential dead zones in rural segments around Pursaiyur and hilly areas.

9. Estimated Emergency Response Times

- **Urban Areas (Salem):** Response times may range from 15 to 30 minutes.
- **Rural Regions:** Could extend to 45 minutes to 1 hour due to distance and road accessibility.

10. Overall Risk Assessment

The route poses moderate risks:

- **Weather and Traffic:** Rains can disrupt travel and lead to accidents due to slick surfaces and poor visibility.
- **Infrastructure:** Rural road quality can impact travel times and vehicle safety.

- Precaution:** Adequate measures, such as monitoring weather forecasts, planning travel to avoid peak traffic, and adhering to safety regulations for hazardous material, can mitigate risks.

Summary

While the route is largely manageable, it requires attention to specific risks tied to weather, traffic congestion, and compliance with safety regulations. Advanced planning is necessary for transporting hazardous materials to ensure safety and efficiency.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63266, 78.12699	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.45358, 77.85781	30 KM/Hr
5	Turn	Medium	11.45353, 77.85697	30 KM/Hr
6	Turn	Medium	11.45839, 77.85143	30 KM/Hr
7	Turn	Medium	11.46307, 77.84941	30 KM/Hr
8	Turn	Medium	11.46316, 77.84921	30 KM/Hr
9	Turn	High	11.45542, 77.81725	15 KM/Hr
10	Blind Spot	Blind Spot	11.45631, 77.81668	10 KM/Hr
11	Blind Spot	Blind Spot	11.62145, 78.10740	10 KM/Hr
12	Turn	High	11.62130, 78.10776	15 KM/Hr
13	Blind Spot	Blind Spot	11.61217, 78.10439	10 KM/Hr
14	Turn	Medium	11.63306, 78.12731	30 KM/Hr
15	Turn	Medium	11.63300, 78.12746	30 KM/Hr
16	Turn	Medium	11.62509, 78.14758	30 KM/Hr
17	Turn	High	11.60280, 78.14689	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693,	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
			78.037627		
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
9	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Thayammal Hospitals	11.6237472, 78.1177631	30 km/h	Medium
14	hospital	Vanitha Hospital	11.6239931, 78.1184622	30 km/h	Medium
16	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
17	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
18	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
19	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
20	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
21	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
22	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
23	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
24	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
25	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
26	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
27	pharmacy	மருந்தகம்	11.6187131, 78.1451978	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
10	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
11	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
12	school	Sri Vidhya Bharati Matric HR Sec School	11.6186069, 78.1103239	30 km/h	Medium
15	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
28	school	Sugam Matric Higher Secondary School	11.6069088, 78.1462056	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63266, 78.12699



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45358, 77.85781



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45839, 77.85143



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46316, 77.84921



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.62145, 78.10740



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.62130, 78.10776



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.61217, 78.10439



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63306, 78.12731



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63300, 78.12746



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62509, 78.14758



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.60280, 78.14689

